Class 12 Mathematics

Chapter 4: Determinants

Practice Set B: step-wise solutions

20 questions **54** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 **MCQ** NCERT Exemplar — 1 mark

There are two values of a which make the determinant $\Delta = \begin{vmatrix} 1 & -2 & 5 \\ 2 & a & -1 \\ 0 & 4 & 2a \end{vmatrix} = 86$. The sum of these two values is:

(A) 4 **(B)** 5 **(C)** -4 **(D)** 9

Formula / idea: Expand a 3×3 determinant along a row: $\Delta = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$.

1 Expand along R_1.
$\Delta = 1(2a^2 + 4) + 2(4a - 0) + 5(8 - 0) = 2a^2 + 8a + 44$

2 Put $\Delta = 86$.
$2a^2 + 8a - 42 = 0 \Rightarrow a^2 + 4a - 21 = 0$

3 Factorise.
$(a + 7)(a - 3) = 0 \Rightarrow a = -7 \text{ or } a = 3$

4 Add the two values. ✓ 1
$-7 + 3 = -4$

5 Hence option (C).

Answer **(C)** -4

Watch out: Taking -2 in place of $+2$ for the middle term: the sign pattern gives $-(-2)(4a) = +8a$.

2 MCQ CBSE SQP 2025-26 1 mark

If A is a square matrix of order 4 and $|\text{adj } A| = 27$, then $A\,(\text{adj } A)$ is equal to:

(A) 3 **(B)** 9 **(C)** $3I$ **(D)** $9I$

Formula / idea: $|\text{adj } A| = |A|^{n-1}$ and $A\,(\text{adj } A) = |A|\,I$.

1 Use $|\text{adj } A| = |A|^{n-1}$ with $n = 4$.
$|A|^3 = 27 \Rightarrow |A| = 3$

2 Use $A\,(\text{adj } A) = |A|\,I$. ✓ 1
$A\,(\text{adj } A) = 3I$

3 The answer is a matrix, not a number, so option (C).

Answer **(C)** $3I$

Watch out: Choosing 3: $A\,(\text{adj } A)$ is a matrix, so the answer must carry I.

3 MCQ CBSE SQP 2023-24 1 mark

Given that A is a square matrix of order 3 and $|A| = -2$, then $|\text{adj}\,(2A)|$ is equal to:

(A) -2^6 **(B)** $+4$ **(C)** -2^8 **(D)** 2^8

Formula / idea: $|kA| = k^n|A|$ and $|\text{adj } B| = |B|^{n-1}$ for order n.

1 Find $|2A|$ with $n = 3$.
$|2A| = 2^3|A| = 8(-2) = -16$

2 $2A$ is also of order 3, so $|\text{adj}\,(2A)| = |2A|^2$. ✓ 1
$|\text{adj}\,(2A)| = (-16)^2 = 256 = 2^8$

3 Hence option (D).

Answer **(D)** 2^8

Watch out: Writing $|2A| = 2|A|$; for order 3 the factor comes out as 2^3.

4 MCQ NCERT Exemplar 1 mark

The area of a triangle with vertices $(-3, 0)$, $(3, 0)$ and $(0, k)$ is 9 sq units. The value of k will be:

(A) 9 **(B)** 3 **(C)** -9 **(D)** 6

Formula / idea: Area $= \frac{1}{2}\left|\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}\right|$

1 Write the area determinant.
$\frac{1}{2}\begin{vmatrix} -3 & 0 & 1 \\ 3 & 0 & 1 \\ 0 & k & 1 \end{vmatrix} = \pm 9$

2 Expand along R_1.
$\frac{1}{2}\left[-3(0-k) - 0 + 1(3k - 0)\right] = \frac{6k}{2} = 3k$

3 Solve. ✓ 1
$3k = \pm 9 \Rightarrow k = \pm 3$

4 Only 3 is among the options, so option (B).

Answer **(B)** 3

Watch out: Forgetting the factor $\frac{1}{2}$ and getting $6k = 9$.

5 MCQ Board-pattern 1 mark

If A_{ij} is the cofactor of the element a_{ij} of the determinant $\Delta = \begin{vmatrix} 1 & 2 & -1 \\ 3 & 0 & 4 \\ -2 & 5 & 1 \end{vmatrix}$, then the value of $a_{11}A_{21} + a_{12}A_{22} + a_{13}A_{23}$ is:

(A) -57 **(B)** 0 **(C)** 57 **(D)** -4

Formula / idea: Elements of one row times the cofactors of another row add up to 0.

1 Cofactors of row 2.
$A_{21} = -(2 + 5) = -7, \quad A_{22} = 1 - 2 = -1, \quad A_{23} = -(5 + 4) = -9$

2 Multiply by the elements of row 1 and add. ✓ 1
$1(-7) + 2(-1) + (-1)(-9) = -7 - 2 + 9 = 0$

3 Row 1 elements with row 2 cofactors give 0, so option (B).

Answer **(B)** 0

Watch out: Using minors instead of cofactors, which gives $7 - 2 - 9 = -4$.

6 **Assertion-Reason** Board-pattern 1 mark

Assertion (A): If A is a square matrix of order 3 with $|A| = -2$, then $|\text{adj}\,(\text{adj}\,A)| = 16$. **Reason (R):** For a square matrix A of order n, $|\text{adj}\,A| = |A|^{n-1}$.

(A) **Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).**

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: $|\text{adj}\,A| = |A|^{n-1}$, and $\text{adj}\,A$ is again a square matrix of order n.

1 Apply R to A (order 3).
$|\text{adj}\,A| = |A|^2 = (-2)^2 = 4$

2 $\text{adj}\,A$ is of order 3, so apply R again.
$|\text{adj}\,(\text{adj}\,A)| = |\text{adj}\,A|^2 = 4^2 = 16$

3 So **Assertion (A) is true.**

4 R is the standard result for the adjoint, so **Reason (R) is true.**

5 A follows by using R twice, so R is the correct explanation of A.

6 Hence option (A). ✓ 1

Answer **(A) Both A and R are true and R is the correct explanation of A.**

Watch out: Thinking the answer must be negative because $|A|$ is negative; even powers remove the sign.

7 **Short answer** PYQ-style 2 marks

Find the values of k if the area of the triangle with vertices $(1, -1)$, $(-4, 2k)$ and $(-k, -5)$ is 24 sq units.

Formula / idea: Area $= \frac{1}{2}\left|\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}\right|$

1 Write the area using determinants. ✓ ½
$\frac{1}{2}\begin{vmatrix} 1 & -1 & 1 \\ -4 & 2k & 1 \\ -k & -5 & 1 \end{vmatrix} = \pm 24$

2 Expand along R_1. ✓ ½
$1(2k + 5) + 1(-4 + k) + 1(20 + 2k^2) = 2k^2 + 3k + 21$

3 Equate to ± 48.
$2k^2 + 3k + 21 = 48 \quad \text{or} \quad 2k^2 + 3k + 21 = -48$

4 First case: factorise. ✓ ½
$2k^2 + 3k - 27 = 0 \Rightarrow (k-3)(2k+9) = 0 \Rightarrow k = 3, -\frac{9}{2}$

5 Second case: $2k^2 + 3k + 69 = 0$ has discriminant $9 - 552 < 0$, so no real k.

6 Hence $k = 3$ or $k = -\frac{9}{2}$. ✓ ½

Answer $k = 3$ **or** $k = -\frac{9}{2}$

Watch out: Expanding $-a_{12}M_{12}$ with the wrong sign: here $-(-1)(-4+k) = -4+k$.

Level 2

17 marks

8 **Short answer** Board-pattern 2 marks

Find the adjoint and the inverse of $A = \begin{bmatrix} a & b \\ c & \frac{1+bc}{a} \end{bmatrix}$, where $a \neq 0$.

Formula / idea: $A^{-1} = \frac{1}{|A|}\operatorname{adj} A$, when $|A| \neq 0$.

1 Find $|A|$. ✓ ½
$|A| = a \cdot \frac{1+bc}{a} - bc = 1 + bc - bc = 1 \neq 0$

2 Swap the diagonal elements and change the signs of the other two. ✓ 1
$\operatorname{adj} A = \begin{bmatrix} \frac{1+bc}{a} & -b \\ -c & a \end{bmatrix}$

3 Divide by $|A| = 1$. ✓ ½
$A^{-1} = \frac{1}{1}\operatorname{adj} A = \begin{bmatrix} \frac{1+bc}{a} & -b \\ -c & a \end{bmatrix}$

Answer $\operatorname{adj} A = A^{-1} = \begin{bmatrix} \frac{1+bc}{a} & -b \\ -c & a \end{bmatrix}$

Watch out: Not simplifying $|A|$ first; it is just 1, so $A^{-1} = \operatorname{adj} A$.

9 **Short answer** NCERT Exemplar 3 marks

Find A^{-1} if $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ and show that $A^{-1} = \dfrac{A^2 - 3I}{2}$.

Formula / idea: $A^{-1} = \dfrac{1}{|A|}\,\text{adj}\,A$, when $|A| \neq 0$.

1 Find $|A|$ by expanding along R_1. ✓ ½

$|A| = 0(0-1) - 1(0-1) + 1(1-0) = 2 \neq 0$

2 Find the cofactors.

$A_{11} = -1, \quad A_{12} = 1, \quad A_{13} = 1$

$A_{21} = 1, \quad A_{22} = -1, \quad A_{23} = 1$

$A_{31} = 1, \quad A_{32} = 1, \quad A_{33} = -1$

3 Transpose the cofactor matrix. ✓ 1

$\text{adj}\,A = \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$

4 Write A^{-1}. ✓ ½

$A^{-1} = \dfrac{1}{2}\begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$

5 Find A^2. ✓ ½

$A^2 = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$

6 Find $\dfrac{A^2 - 3I}{2}$.

$\dfrac{A^2 - 3I}{2} = \dfrac{1}{2}\begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$

7 This equals A^{-1}, hence $A^{-1} = \dfrac{A^2 - 3I}{2}$. ✓ ½

Answer $A^{-1} = \dfrac{1}{2}\begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix} = \dfrac{A^2 - 3I}{2}$

Watch out: Subtracting 3 from every element of A^2; $3I$ changes only the diagonal.

10 **Short answer** Board-pattern 2 marks

Using determinants, find the equation of the line joining $A(2,-3)$ and $B(-1,4)$. Hence find k if the point $C(k,-10)$ lies on this line.

Formula / idea: $P(x,y)$ lies on AB when the area of $\triangle PAB$ is 0.

1. Let $P(x,y)$ be any point on AB. Area of $\triangle PAB = 0$.
$$\begin{vmatrix} x & y & 1 \\ 2 & -3 & 1 \\ -1 & 4 & 1 \end{vmatrix} = 0$$ ✓ ½

2. Expand along R_1.
$x(-3-4) - y(2+1) + 1(8-3) = 0 \;\Rightarrow\; -7x - 3y + 5 = 0$ ✓ ½

3. So the line is $7x + 3y - 5 = 0$. ✓ ½

4. Put $C(k,-10)$ in the equation.
$7k - 30 - 5 = 0 \;\Rightarrow\; k = 5$ ✓ ½

Answer $7x + 3y - 5 = 0$; $k = 5$

Watch out: Writing the middle term as $+y(2+1)$; the R_1 expansion has $-y$.

11 **Short answer** CBSE 2023 3 marks

If $A = \begin{bmatrix} 3 & 2 \\ 5 & -7 \end{bmatrix}$, then find A^{-1} and use it to solve the following system of equations: $3x + 5y = 11$, $2x - 7y = -3$.

Formula / idea: The coefficient matrix here is A', and $(A')^{-1} = (A^{-1})'$.

1. Find $|A|$.
$|A| = -21 - 10 = -31 \neq 0$ ✓ ½

2. Find adj A.
$\text{adj}\, A = \begin{bmatrix} -7 & -2 \\ -5 & 3 \end{bmatrix}$ ✓ ½

3. Write A^{-1}.
$A^{-1} = -\frac{1}{31}\begin{bmatrix} -7 & -2 \\ -5 & 3 \end{bmatrix}$ ✓ ½

4. The columns of A are the rows of the system, so it reads $A'X = B$.
$\begin{bmatrix} 3 & 5 \\ 2 & -7 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 11 \\ -3 \end{bmatrix}$ ✓ ½

5. So $X = (A')^{-1}B = (A^{-1})'B$.
$X = -\frac{1}{31}\begin{bmatrix} -7 & -5 \\ -2 & 3 \end{bmatrix}\begin{bmatrix} 11 \\ -3 \end{bmatrix} = -\frac{1}{31}\begin{bmatrix} -62 \\ -31 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ ✓ ½

6 Hence $x = 2, y = 1$. ✓ ½

Answer $A^{-1} = -\frac{1}{31}\begin{bmatrix} -7 & -2 \\ -5 & 3 \end{bmatrix}$; $x = 2, y = 1$

Watch out: Using $X = A^{-1}B$ directly; the coefficient matrix is A', not A.

12 **Short answer** Board-pattern 2 marks

Using the matrix method, find the value of k for which the system of equations $x + 2y = 3$, $3x + ky = 5$ has no solution.

Formula / idea: If $|A| = 0$ and $(\text{adj } A)B \neq O$, the system is inconsistent.

1 Write as $AX = B$ and find $|A|$. ✓ ½
$A = \begin{bmatrix} 1 & 2 \\ 3 & k \end{bmatrix}$, $B = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$, $|A| = k - 6$

2 If $|A| \neq 0$ the solution is unique, so we need $|A| = 0$. ✓ ½
$k - 6 = 0 \Rightarrow k = 6$

3 For $k = 6$, find $\text{adj } A$.
$\text{adj } A = \begin{bmatrix} 6 & -2 \\ -3 & 1 \end{bmatrix}$

4 Find $(\text{adj } A)B$. ✓ ½
$(\text{adj } A)B = \begin{bmatrix} 18 - 10 \\ -9 + 5 \end{bmatrix} = \begin{bmatrix} 8 \\ -4 \end{bmatrix} \neq O$

5 So the system is inconsistent, hence it has no solution when $k = 6$. ✓ ½

Answer $k = 6$

Watch out: Stopping at $|A| = 0$; you must also show $(\text{adj } A)B \neq O$ to rule out infinitely many solutions.

13 **Short answer** PYQ-style 2 marks

If $A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$, show that $A'A^{-1} = \begin{bmatrix} \cos 2x & -\sin 2x \\ \sin 2x & \cos 2x \end{bmatrix}$.

Formula / idea: $\cos 2x = \cos^2 x - \sin^2 x$ and $\sin 2x = 2 \sin x \cos x$.

1 Find $|A|$.
$|A| = 1 + \tan^2 x = \sec^2 x \neq 0$

2 Find $A^{-1} = \frac{1}{|A|}\text{adj } A$. ✓ ½
$A^{-1} = \cos^2 x \begin{bmatrix} 1 & -\tan x \\ \tan x & 1 \end{bmatrix}$

3 Write A'; it equals adj A here.

$$A' = \begin{bmatrix} 1 & -\tan x \\ \tan x & 1 \end{bmatrix}$$

4 Multiply. ✓ ½

$$A'A^{-1} = \cos^2 x \begin{bmatrix} 1-\tan^2 x & -2\tan x \\ 2\tan x & 1-\tan^2 x \end{bmatrix}$$

5 Take $\cos^2 x$ inside. ✓ ½

$$= \begin{bmatrix} \cos^2 x - \sin^2 x & -2\sin x\cos x \\ 2\sin x\cos x & \cos^2 x - \sin^2 x \end{bmatrix}$$

6 Use double-angle formulas. ✓ ½

$$A'A^{-1} = \begin{bmatrix} \cos 2x & -\sin 2x \\ \sin 2x & \cos 2x \end{bmatrix}$$

Answer $A'A^{-1} = \begin{bmatrix} \cos 2x & -\sin 2x \\ \sin 2x & \cos 2x \end{bmatrix}$. **Shown.**

Watch out: Writing $|A| = 1 - \tan^2 x$; here $bc = (\tan x)(-\tan x)$, so $ad - bc = 1 + \tan^2 x$.

14 **Short answer** Board-pattern 3 marks

If $A^{-1} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, find $(AB)^{-1}$.

Formula / idea: Reversal law: $(AB)^{-1} = B^{-1}A^{-1}$.

1 Find $|B|$. ✓ ½

$$|B| = 4 - 6 = -2 \neq 0$$

2 Find $B^{-1} = \dfrac{1}{|B|}\,\text{adj}\, B$. ✓ ½

$$B^{-1} = -\frac{1}{2}\begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$$

3 Use the reversal law. ✓ ½

$$(AB)^{-1} = B^{-1}A^{-1} = -\frac{1}{2}\begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}\begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$$

4 Multiply. ✓ 1

$$= -\frac{1}{2}\begin{bmatrix} 12+10 & -4-4 \\ -9-5 & 3+2 \end{bmatrix} = -\frac{1}{2}\begin{bmatrix} 22 & -8 \\ -14 & 5 \end{bmatrix}$$

5 Hence $(AB)^{-1} = \begin{bmatrix} -11 & 4 \\ 7 & -\frac{5}{2} \end{bmatrix}$. ✓ ½

Answer $(AB)^{-1} = -\frac{1}{2}\begin{bmatrix} 22 & -8 \\ -14 & 5 \end{bmatrix} = \begin{bmatrix} -11 & 4 \\ 7 & -\frac{5}{2} \end{bmatrix}$

Watch out: Multiplying in the order $A^{-1}B^{-1}$; the reversal law needs $B^{-1}A^{-1}$.

Level 3

29 marks

15 **Long answer** PYQ-style 5 marks

A school wants to award its students for the values of Honesty, Regularity and Hard work with a total cash award of ₹6,000. Three times the award money for Hard work added to that given for Honesty amounts to ₹11,000. The award money given for Honesty and Hard work together is double the one given for Regularity. Represent the situation algebraically and find the award money for each value, using the matrix method.

Formula / idea: $AX = B \Rightarrow X = A^{-1}B$, where $A^{-1} = \frac{1}{|A|}\operatorname{adj} A$.

1 Let the awards for Honesty, Regularity, Hard work be ₹x, ₹y, ₹z. ✓ 1
$x + y + z = 6000, \quad x + 3z = 11000, \quad x - 2y + z = 0$

2 Write as $AX = B$.
$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 3 \\ 1 & -2 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6000 \\ 11000 \\ 0 \end{bmatrix}$

3 Find $|A|$ by expanding along R_1. ✓ ½
$|A| = 1(0+6) - 1(1-3) + 1(-2-0) = 6 \neq 0$

4 Cofactors of row 1.
$A_{11} = 6, \quad A_{12} = 2, \quad A_{13} = -2$

5 Cofactors of row 2.
$A_{21} = -3, \quad A_{22} = 0, \quad A_{23} = 3$

6 Cofactors of row 3. ✓ 1
$A_{31} = 3, \quad A_{32} = -2, \quad A_{33} = -1$

7 $\operatorname{adj} A$ is the transpose of the cofactor matrix. ✓ ½
$\operatorname{adj} A = \begin{bmatrix} 6 & -3 & 3 \\ 2 & 0 & -2 \\ -2 & 3 & -1 \end{bmatrix}$

8 Write A^{-1}.

$$A^{-1} = \frac{1}{6}\begin{bmatrix} 6 & -3 & 3 \\ 2 & 0 & -2 \\ -2 & 3 & -1 \end{bmatrix}$$

✓ ½

9 Find $X = A^{-1}B$.

$$X = \frac{1}{6}\begin{bmatrix} 36000 - 33000 + 0 \\ 12000 + 0 - 0 \\ -12000 + 33000 - 0 \end{bmatrix} = \frac{1}{6}\begin{bmatrix} 3000 \\ 12000 \\ 21000 \end{bmatrix} = \begin{bmatrix} 500 \\ 2000 \\ 3500 \end{bmatrix}$$

✓ 1

10 Hence Honesty ₹500, Regularity ₹2,000 and Hard work ₹3,500.

✓ ½

Answer **Honesty ₹500, Regularity ₹2,000, Hard work ₹3,500**

Watch out: Writing the third condition as $2(x + z) = y$; it is $x + z = 2y$.

16 Long answer CBSE 2025 5 marks

Given $A = \begin{bmatrix} -4 & 4 & 4 \\ -7 & 1 & 3 \\ 5 & -3 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & -2 \\ 2 & 1 & 3 \end{bmatrix}$, find AB. Hence, solve the system of linear equations: $x - y + z = 4$, $x - 2y - 2z = 9$, $2x + y + 3z = 1$.

Formula / idea: If $AB = kI$ with $k \neq 0$, then $B^{-1} = \frac{1}{k}A$.

1 Row 1 of AB.

$(-4 + 4 + 8,\ 4 - 8 + 4,\ -4 - 8 + 12) = (8,\ 0,\ 0)$

✓ ½

2 Row 2 of AB.

$(-7 + 1 + 6,\ 7 - 2 + 3,\ -7 - 2 + 9) = (0,\ 8,\ 0)$

✓ ½

3 Row 3 of AB.

$(5 - 3 - 2,\ -5 + 6 - 1,\ 5 + 6 - 3) = (0,\ 0,\ 8)$

✓ ½

4 So AB is 8 times the identity.

$$AB = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 8 \end{bmatrix} = 8I$$

✓ ½

5 The system has coefficient matrix B: $BX = C$.

$$\begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & -2 \\ 2 & 1 & 3 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 9 \\ 1 \end{bmatrix}$$

✓ ½

6 From $AB = 8I$: $\left(\frac{1}{8}A\right)B = I$.

$$B^{-1} = \frac{1}{8}A$$

✓ 1

7 Find $X = B^{-1}C$.

$$X = \frac{1}{8}\begin{bmatrix}-4 & 4 & 4\\ -7 & 1 & 3\\ 5 & -3 & -1\end{bmatrix}\begin{bmatrix}4\\9\\1\end{bmatrix} = \frac{1}{8}\begin{bmatrix}-16+36+4\\ -28+9+3\\ 20-27-1\end{bmatrix}$$

✓ ½

8 Simplify.

$$X = \frac{1}{8}\begin{bmatrix}24\\-16\\-8\end{bmatrix} = \begin{bmatrix}3\\-2\\-1\end{bmatrix}$$

✓ ½

9 Hence $x = 3, y = -2, z = -1$.

✓ ½

Answer $AB = 8I$; $x = 3, y = -2, z = -1$

Watch out: Taking $B^{-1} = A$; since $AB = 8I$, not I, divide by 8.

17 **Long answer** NCERT Misc. Ex. Q7 5 marks

Solve the following system of equations by the matrix method: $\frac{2}{x} + \frac{3}{y} + \frac{10}{z} = 4$, $\frac{4}{x} - \frac{6}{y} + \frac{5}{z} = 1$, $\frac{6}{x} + \frac{9}{y} - \frac{20}{z} = 2$.

Formula / idea: Put $\frac{1}{x} = u$, $\frac{1}{y} = v$, $\frac{1}{z} = w$ to get a linear system; then $X = A^{-1}B$.

1 Put $\frac{1}{x} = u$, $\frac{1}{y} = v$, $\frac{1}{z} = w$ and write as $AX = B$.

$$\begin{bmatrix}2 & 3 & 10\\ 4 & -6 & 5\\ 6 & 9 & -20\end{bmatrix}\begin{bmatrix}u\\v\\w\end{bmatrix} = \begin{bmatrix}4\\1\\2\end{bmatrix}$$

✓ ½

2 Find $|A|$ by expanding along R_1.

$|A| = 2(120-45) - 3(-80-30) + 10(36+36) = 150 + 330 + 720 = 1200 \neq 0$

✓ 1

3 Cofactors of row 1.

$A_{11} = 75, \quad A_{12} = 110, \quad A_{13} = 72$

✓ ½

4 Cofactors of row 2.

$A_{21} = 150, \quad A_{22} = -100, \quad A_{23} = 0$

✓ ½

5 Cofactors of row 3.

$A_{31} = 75, \quad A_{32} = 30, \quad A_{33} = -24$

✓ ½

6 $\operatorname{adj} A$ is the transpose of the cofactor matrix.

$$\operatorname{adj} A = \begin{bmatrix}75 & 150 & 75\\ 110 & -100 & 30\\ 72 & 0 & -24\end{bmatrix}$$

✓ ½

7 Write $A^{-1} = \frac{1}{|A|}\operatorname{adj} A$.

$$A^{-1} = \frac{1}{1200}\begin{bmatrix} 75 & 150 & 75 \\ 110 & -100 & 30 \\ 72 & 0 & -24 \end{bmatrix}$$

✓ ½

8 Find $X = A^{-1}B$.

$$\begin{bmatrix} u \\ v \\ w \end{bmatrix} = \frac{1}{1200}\begin{bmatrix} 300+150+150 \\ 440-100+60 \\ 288+0-48 \end{bmatrix} = \frac{1}{1200}\begin{bmatrix} 600 \\ 400 \\ 240 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{3} \\ \frac{1}{5} \end{bmatrix}$$

✓ ½

9 Go back to x, y, z.

$$\frac{1}{x} = \frac{1}{2}, \quad \frac{1}{y} = \frac{1}{3}, \quad \frac{1}{z} = \frac{1}{5}$$

10 Hence $x = 2, y = 3, z = 5$.

✓ ½

Answer $x = 2,\ y = 3,\ z = 5$

Watch out: Stopping at $u = \frac{1}{2}$ and writing $x = \frac{1}{2}$; u is $\frac{1}{x}$, so $x = 2$.

18 Long answer Board-pattern 5 marks

Ritu invests a total of ₹10,000 in three schemes P, Q and R, which pay simple interest at 5%, 8% and 9% per annum respectively. Her total interest for one year is ₹770. The amount invested in scheme Q is equal to the sum of the amounts invested in P and R. Using the matrix method, find the amount invested in each scheme.

Formula / idea: $AX = B \Rightarrow X = A^{-1}B$, where $A^{-1} = \frac{1}{|A|}\operatorname{adj} A$.

1 Let the amounts in P, Q, R be ₹x, ₹y, ₹z.

$$x + y + z = 10000, \quad \frac{5x}{100} + \frac{8y}{100} + \frac{9z}{100} = 770, \quad y = x + z$$

2 Clear fractions and rearrange.

$$x + y + z = 10000, \quad 5x + 8y + 9z = 77000, \quad x - y + z = 0$$

✓ 1

3 Write as $AX = B$.

$$\begin{bmatrix} 1 & 1 & 1 \\ 5 & 8 & 9 \\ 1 & -1 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10000 \\ 77000 \\ 0 \end{bmatrix}$$

4 Find $|A|$ by expanding along R_1.

$$|A| = 1(8+9) - 1(5-9) + 1(-5-8) = 17 + 4 - 13 = 8 \neq 0$$

✓ ½

5 Cofactors of row 1.

$$A_{11} = 17, \quad A_{12} = 4, \quad A_{13} = -13$$

6 Cofactors of row 2.
$A_{21} = -2, \quad A_{22} = 0, \quad A_{23} = 2$

7 Cofactors of row 3. ✓ 1
$A_{31} = 1, \quad A_{32} = -4, \quad A_{33} = 3$

8 $\text{adj}\, A$ is the transpose of the cofactor matrix. ✓ ½
$$\text{adj}\, A = \begin{bmatrix} 17 & -2 & 1 \\ 4 & 0 & -4 \\ -13 & 2 & 3 \end{bmatrix}$$

9 Write $A^{-1} = \frac{1}{|A|}\,\text{adj}\, A$. ✓ ½
$$A^{-1} = \frac{1}{8}\begin{bmatrix} 17 & -2 & 1 \\ 4 & 0 & -4 \\ -13 & 2 & 3 \end{bmatrix}$$

10 Find $X = A^{-1}B$. ✓ 1
$$X = \frac{1}{8}\begin{bmatrix} 170000 - 154000 + 0 \\ 40000 + 0 - 0 \\ -130000 + 154000 + 0 \end{bmatrix} = \frac{1}{8}\begin{bmatrix} 16000 \\ 40000 \\ 24000 \end{bmatrix} = \begin{bmatrix} 2000 \\ 5000 \\ 3000 \end{bmatrix}$$

11 Hence ₹2,000 in P, ₹5,000 in Q and ₹3,000 in R. ✓ ½

Answer **P: ₹2,000, Q: ₹5,000, R: ₹3,000**

Watch out: Writing the interest equation as $5x + 8y + 9z = 770$; multiplying by 100 gives 77000.

19 Long answer PYQ-style 5 marks

If $A = \begin{bmatrix} 2 & 1 & -3 \\ 3 & 2 & 1 \\ 1 & 2 & -1 \end{bmatrix}$, find A^{-1}. Using A^{-1}, solve the system of equations: $2x + 3y + z = 4$, $x + 2y + 2z = 6$, $-3x + y - z = -10$.

Formula / idea: The coefficient matrix here is A', and $(A')^{-1} = (A^{-1})'$.

1 Find $|A|$ by expanding along R_1. ✓ ½
$|A| = 2(-2 - 2) - 1(-3 - 1) - 3(6 - 2) = -8 + 4 - 12 = -16 \neq 0$

2 Cofactors of row 1. ✓ ½
$A_{11} = -4, \quad A_{12} = 4, \quad A_{13} = 4$

3 Cofactors of row 2. ✓ ½
$A_{21} = -5, \quad A_{22} = 1, \quad A_{23} = -3$

4 Cofactors of row 3. ✓ ½
$A_{31} = 7, \quad A_{32} = -11, \quad A_{33} = 1$

5 $\operatorname{adj} A$ is the transpose of the cofactor matrix. ✓ ½

$$\operatorname{adj} A = \begin{bmatrix} -4 & -5 & 7 \\ 4 & 1 & -11 \\ 4 & -3 & 1 \end{bmatrix}$$

6 Write A^{-1}. ✓ ½

$$A^{-1} = -\frac{1}{16}\begin{bmatrix} -4 & -5 & 7 \\ 4 & 1 & -11 \\ 4 & -3 & 1 \end{bmatrix}$$

7 The columns of A are the rows of the system, so it reads $A'X = B$. ✓ ½

$$A'X = B, \quad B = \begin{bmatrix} 4 \\ 6 \\ -10 \end{bmatrix}$$

8 So $X = (A')^{-1}B = (A^{-1})'B$. ✓ ½

$$X = -\frac{1}{16}\begin{bmatrix} -4 & 4 & 4 \\ -5 & 1 & -3 \\ 7 & -11 & 1 \end{bmatrix}\begin{bmatrix} 4 \\ 6 \\ -10 \end{bmatrix}$$

9 Multiply. ✓ ½

$$X = -\frac{1}{16}\begin{bmatrix} -16+24-40 \\ -20+6+30 \\ 28-66-10 \end{bmatrix} = -\frac{1}{16}\begin{bmatrix} -32 \\ 16 \\ -48 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$$

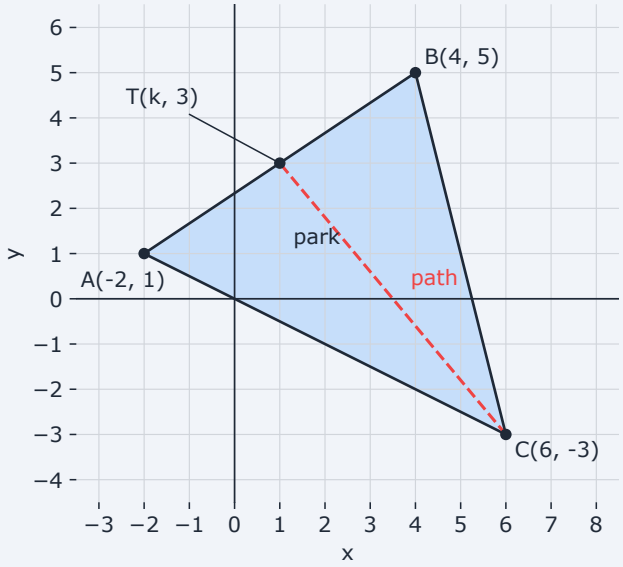
10 Hence $x = 2, y = -1, z = 3$. ✓ ½

Answer $A^{-1} = -\frac{1}{16}\begin{bmatrix} -4 & -5 & 7 \\ 4 & 1 & -11 \\ 4 & -3 & 1 \end{bmatrix}$; $x = 2$, $y = -1$, $z = 3$

Watch out: Using $X = A^{-1}B$; here the first equation has coefficients $2, 3, 1$, the first column of A.

20 **Case study** Board-pattern 4 marks

Case Study: A town park is in the shape of a triangle with corners $A(-2, 1)$, $B(4, 5)$ and $C(6, -3)$ on a map grid (1 unit $= 10$ m). A water tap $T(k, 3)$ is fixed on the fence AB, and a straight path is to be laid from the gate at C to the tap T.



(i) Using determinants, find the area of the park in square metres. 1 mark

(ii) Using determinants, find the value of k. 1 mark

(iii) Using determinants, find the equation of the path CT. 2 marks

Formula / idea: Area $= \frac{1}{2}\left|\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}\right|$; area 0 means collinear.

(i) 1 mark

1 Area of $\triangle ABC$ in square units. ✓ ½

$$\Delta = \frac{1}{2}\left|\begin{vmatrix} -2 & 1 & 1 \\ 4 & 5 & 1 \\ 6 & -3 & 1 \end{vmatrix}\right| = \frac{1}{2}\,|-2(8) - 1(-2) + 1(-42)| = \frac{56}{2} = 28$$

2 1 sq unit $= 10 \times 10 = 100$ m^2. ✓ ½

$$\text{Area} = 28 \times 100 = 2800 \text{ m}^2$$

Answer (i) 2800 **m**^{2}

(ii) 1 mark

1 T lies on AB, so the area of $\triangle ABT$ is 0. ✓ ½

$$\begin{vmatrix} -2 & 1 & 1 \\ 4 & 5 & 1 \\ k & 3 & 1 \end{vmatrix} = 0$$

2	Expand along R_1 and solve. $-2(5-3)-1(4-k)+1(12-5k)=4-4k=0 \Rightarrow k=1$	✓ ½

Answer (ii) $k=1$

(iii)

2 marks

1	Let $P(x,y)$ be any point on CT, with $C(6,-3)$ and $T(1,3)$. $\begin{vmatrix} x & y & 1 \\ 6 & -3 & 1 \\ 1 & 3 & 1 \end{vmatrix} = 0$	✓ ½
2	Expand along R_1. $x(-3-3)-y(6-1)+1(18+3)=0$	✓ ½
3	Simplify. $-6x-5y+21=0$	✓ ½
4	Required path: $6x+5y-21=0$.	✓ ½

Answer (iii) $6x+5y-21=0$

Watch out: Giving the area as $28\ \text{m}^2$; each square unit is $10\ \text{m} \times 10\ \text{m} = 100\ \text{m}^2$.